

CS2601 Linear and Convex Optimization

Homework 3

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1. (a.) Suppose $\mathbf{x}, \mathbf{y} \in S, \forall \theta \in [0, 1]$. Let $f(x) = x \log x, x > 0$
 $\therefore f'(x) = 1 + \log x, f''(x) = \frac{1}{x} > 0$
 $\therefore f(x)$ is a convex function
 $\therefore (\theta x_1 + \bar{\theta} y_1) \log(\theta x_1 + \bar{\theta} y_1) + (\theta x_2 + \bar{\theta} y_2) \log(\theta x_2 + \bar{\theta} y_2) \leq (\theta x_1 \log x_1 + \bar{\theta} y_1 \log y_1) + (\theta x_2 \log x_2 + \bar{\theta} y_2 \log y_2) = \theta(x_1 \log x_1 + x_2 \log x_2) + \bar{\theta}(y_1 \log y_1 + y_2 \log y_2) \leq 2$
 $\therefore \theta \mathbf{x} + \bar{\theta} \mathbf{y} \in S$
 $\therefore S$ is a convex set
 (b.) Let $g_1(\mathbf{x}) = \|\mathbf{Ax} + \mathbf{b}\|^2, h_1(x) = x^{\frac{3}{2}}$, Then $g_1(\mathbf{x})$ is convex, $h_1'(x) = \frac{3}{2}x^{\frac{1}{2}} > 0, h_1''(x) = \frac{3}{4}x^{-\frac{1}{2}} > 0$
 $\therefore h_1(x)$ is convex and increasing.
 $\therefore f_1(\mathbf{x}) = \|\mathbf{Ax} + \mathbf{b}\|^3 = h_1(g_1(\mathbf{x}))$ is convex
 Let $g_2(\mathbf{x}) = 5x_1 + 3x_2, h_2(x) = \log(1 + e^x)$, Then $\nabla^2 g_2(\mathbf{x}) = \mathbf{O}, g_2(\mathbf{x})$ is convex, $h_2'(x) = \frac{e^x}{1+e^x} > 0, h_2''(x) = \frac{e^{-x}}{(1+e^{-x})^2} > 0$
 $\therefore h_2(x)$ is convex and increasing.
 $\therefore f_2(\mathbf{x}) = \log(1 + e^{5x_1+3x_2}) = h_2(g_2(\mathbf{x}))$ is convex
 $\therefore f(\mathbf{x}) = \max\{f_1(\mathbf{x}), f_2(\mathbf{x})\}$ is convex.
 $\therefore \forall \mathbf{x}, \mathbf{y} \in S, \theta \in [0, 1]$
 $f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y}) \leq 2$
 $\therefore \theta \mathbf{x} + \bar{\theta} \mathbf{y} \in S, S$ is convex.

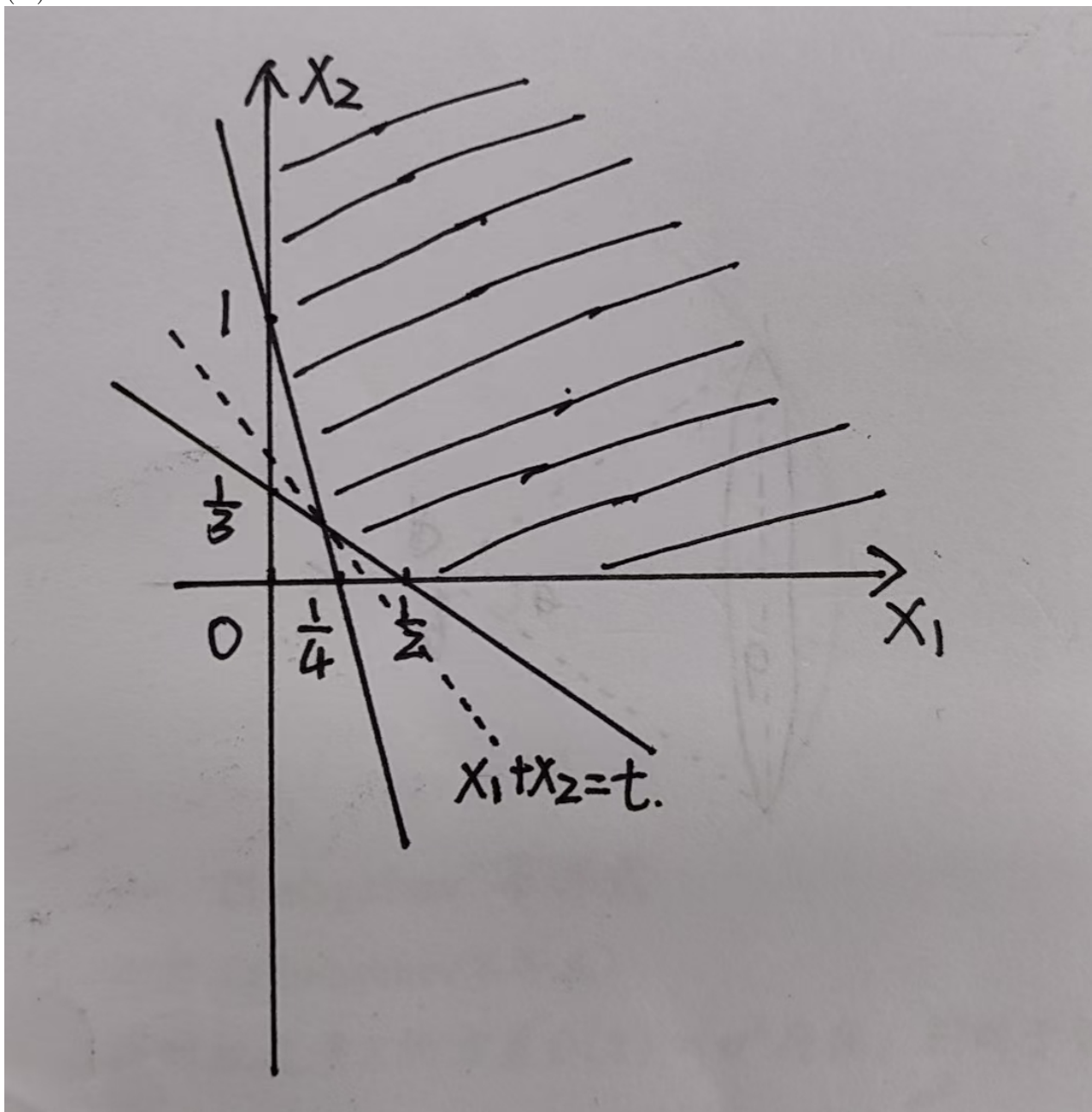
2. $\therefore f(x) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) = \frac{1}{2}(\mathbf{x}^T \mathbf{x} - \mathbf{x}_0^T \mathbf{x} - \mathbf{x}^T \mathbf{x}_0 + \mathbf{x}_0^T \mathbf{x}_0), \mathbf{x}^T \mathbf{x}_0 \in R$
 $\therefore f(x) = \frac{1}{2}(\mathbf{x}^T \mathbf{x} - 2\mathbf{x}_0^T \mathbf{x} + \mathbf{x}_0^T \mathbf{x}_0)$
 Then, we consider the constraint $\|\mathbf{x}\| \leq 1$, that is $\mathbf{x}^T \mathbf{x} \leq 1$
 Define the Lagrangian function $L(\mathbf{x}, \lambda) = \frac{1}{2}(\mathbf{x}^T \mathbf{x} - 2\mathbf{x}_0^T \mathbf{x} + \mathbf{x}_0^T \mathbf{x}_0) + \lambda(\mathbf{x}^T \mathbf{x} - 1), \lambda > 0$

$$\begin{cases} \mathbf{x} - \mathbf{x}_0 + 2\lambda \mathbf{x} = 0 \\ \mathbf{x}^T \mathbf{x} - 1 \leq 0 \\ \lambda \geq 0 \\ \lambda(\mathbf{x}^T \mathbf{x} - 1) = 0 \end{cases}$$
 Then $\mathbf{x}^* = \frac{\mathbf{x}_0}{\|\mathbf{x}_0\|}$

3. (a.) No. Because $h(\mathbf{x}) = x_1^2 - 3x_2$ is not an affine function
 (b.) Let $f(\mathbf{x}) = x_1^2 + x_2^4, g_1(\mathbf{x}) = (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2}, g_2(\mathbf{x}) = x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2, h(\mathbf{x}) = 6x_1 - 7x_2$
 $\therefore \nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 12x_2^2 \end{bmatrix} \succeq \mathbf{O}, f(\mathbf{x})$ is convex
 $\nabla^2 g_1(\mathbf{x}) = \begin{bmatrix} 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \\ 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \end{bmatrix} \succeq \mathbf{O}, g_1(\mathbf{x})$ is convex
 $\nabla^2 g_2(\mathbf{x}) = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \succeq \mathbf{O}, g_2(\mathbf{x})$ is convex

$h(\mathbf{x}) = [6, -7]\mathbf{x}$, $h(\mathbf{x})$ is an affine function
 $\therefore$ It is a convex optimization problem.

4. (a.)



Suppose the optimal solution is $\mathbf{x}^*$. We can see from the picture that

$$\begin{cases} 4x_1^* + x_2^* = 1 \\ 2x_1^* + 3x_2^* = 1 \end{cases} \therefore \mathbf{x}^* = (0.2, 0.2), \text{ and the optimal value is } f(\mathbf{x}^*) = 0.4$$

(b.) From the picture above we can know that $x_1 + x_2$ has no maximum value.

$\therefore f(x_1, x_2) = -x_1 - x_2$ has no optimal value nor optimal solution

5. (a.) Because $\nabla f(\mathbf{x}) = \mathbf{Q}\mathbf{x}$
 $\therefore \lambda_1(\mathbf{Q}) = \gamma, \lambda_2(\mathbf{Q}) = 1$

If $\gamma \geq 1, L_{min} = \gamma$; If $\gamma < 1, L_{min} = 1$

(b.)Outputs:

```
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.1, stepsize=2.2, number of iterations=10
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.1, stepsize=1, number of iterations=88
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.1, stepsize=0.5, number of iterations=180
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.1, stepsize=0.01, number of iterations=9206
PS D:\线性优化与凸优化\code_h3> █
```

When stepsize = 2.2, doesn't coverage

When stepsize = 1, coverage. Number of iterations=88

When stepsize = 0.5, coverage. Number of iterations = 180

When stepsize = 0.01, coverage. Number of iterations = 9206

(Please see next page for plots)

(c.)Outputs:

```
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=1, stepsize=1, number of iterations=1
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.1, stepsize=1, number of iterations=88
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.01, stepsize=1, number of iterations=688
PS D:\线性优化与凸优化\code_h3> python -u "d:\线性优化与凸优化\code_h3\p5.py"
gamma=0.001, stepsize=1, number of iterations=4603
```

When $\gamma = 1$, number of iterations = 1

When $\gamma = 0.1$, number of iterations = 88

When $\gamma = 0.01$, number of iterations = 688

When $\gamma = 0.001$, number of iterations = 4603

As γ decreases, the number of iterations increases

6. (a.) $f(\mathbf{w}) = (\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y}) = \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{y}^T \mathbf{X} \mathbf{w} + \|\mathbf{y}\|_2^2$

$\therefore \nabla f(\mathbf{w}) = 2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y}$

Let $\nabla f(\mathbf{w}) = 0$, then $\mathbf{w}^* = (2, 1)^T$, $f(\mathbf{w})_{min} = 4$

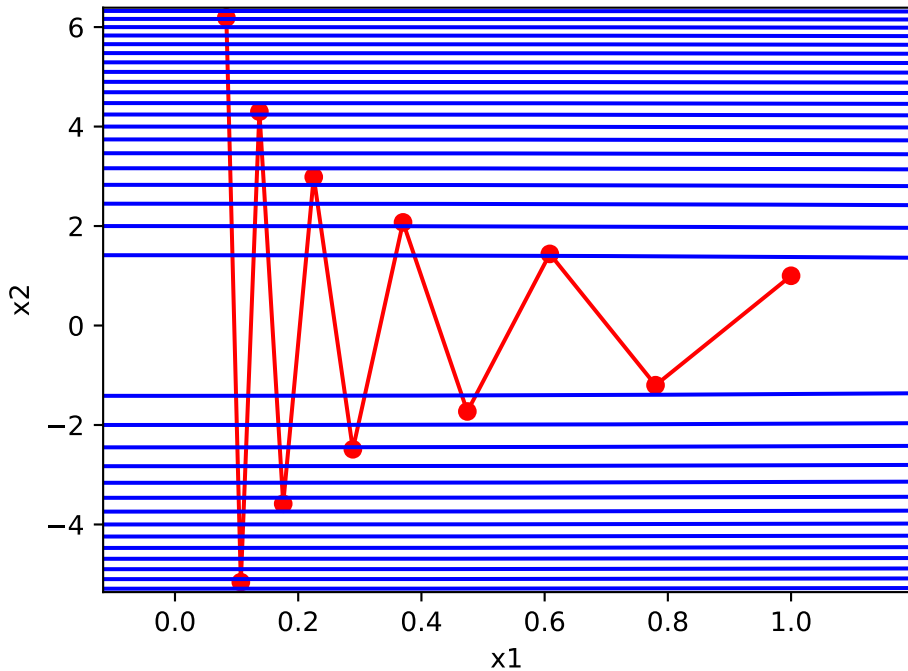
(b.)Choose $\mathbf{w}_0 = (1, 2)^T$, stepsize = 0.1.

Outputs:

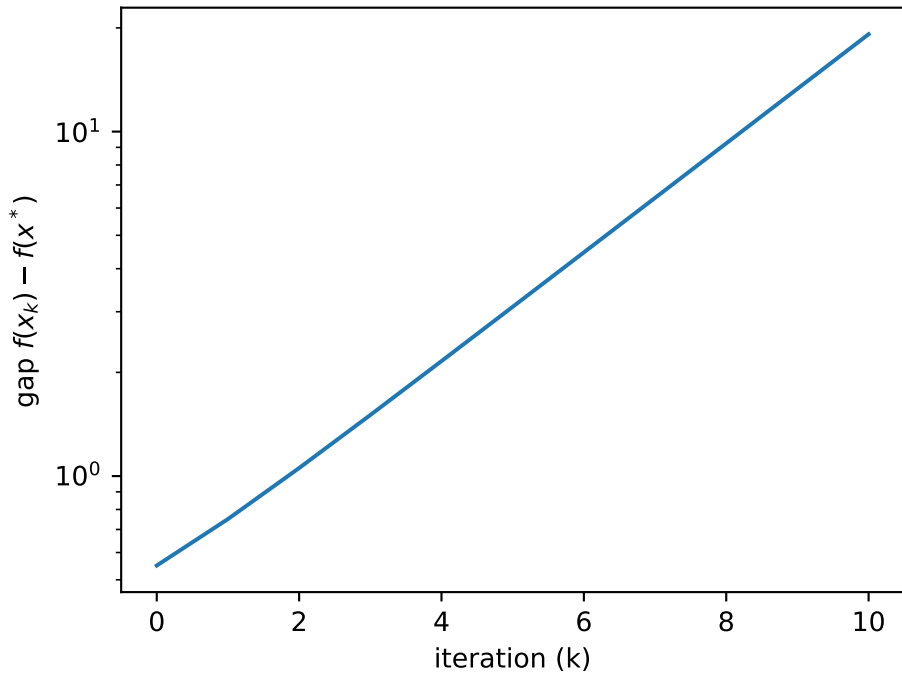
```
PS D:\线性优化与凸优化\code_h3\Problem 6> python -u "d:\线性优化与凸优化\code_h3\Problem 6\p5.py"
optimal solution: [[2.
 1.00000468]]
optimal value: [[4.]]
stepsize=0.1, number of iterations=55
PS D:\线性优化与凸优化\code_h3\Problem 6> █
```

The two solutions agree with each other.

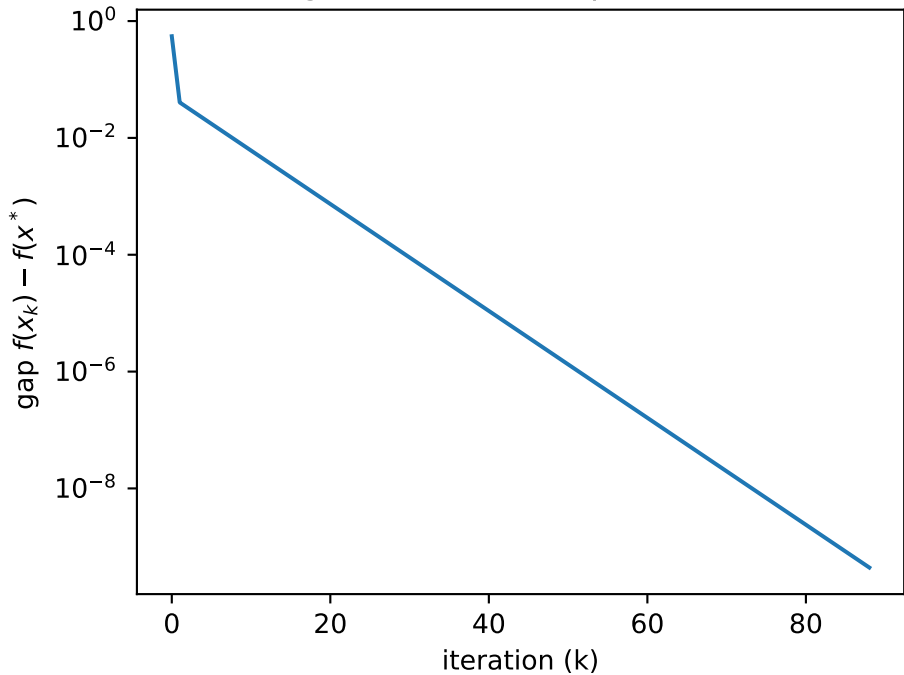
$\gamma = 0.1$, stepsize = 2.2



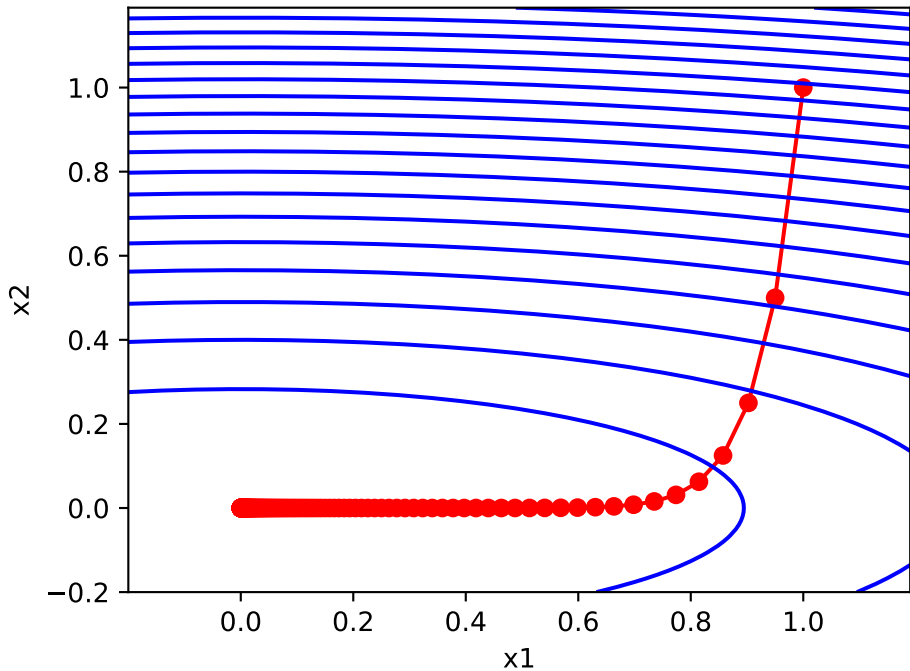
gamma = 0.1, stepsize = 2.2



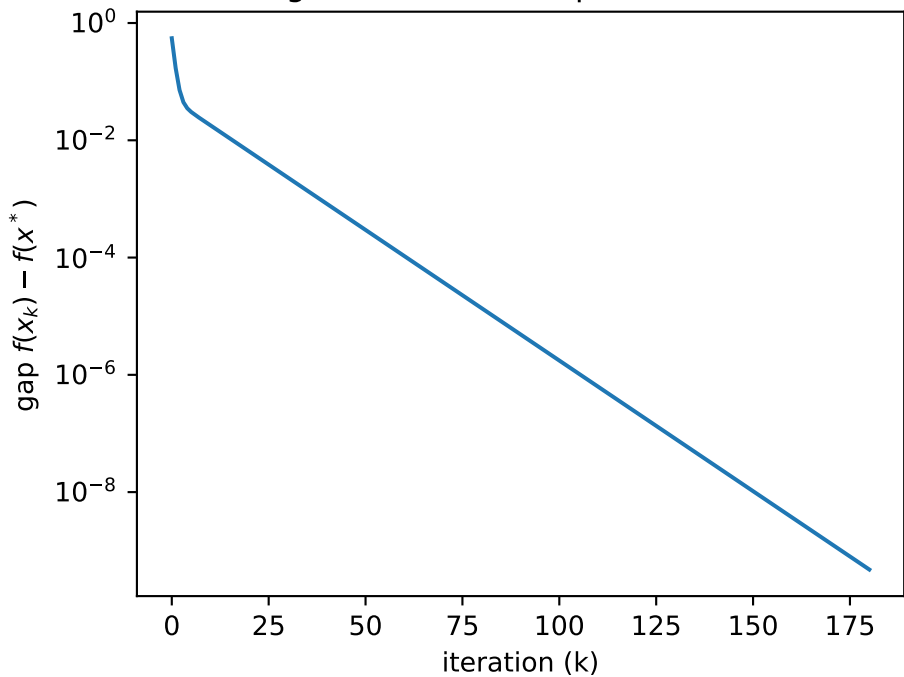
gamma = 0.1, stepsize = 1



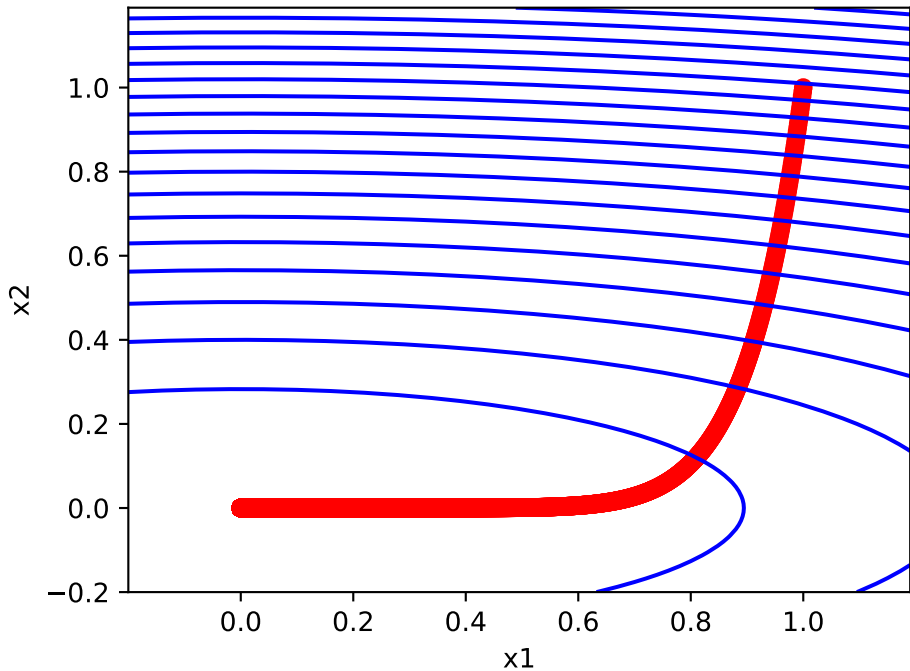
$\gamma = 0.1$, stepsize = 0.5



gamma = 0.1, stepsize = 0.5



$\gamma = 0.1$, stepsize = 0.01



gamma = 0.01, stepsize = 1

